
xibili
西比里科技
Xibili Technology
重构能源脉动
Reimagining the Pulse of Energy



公司概况
Corporate Profile

西比里科技由国家级高新技术企业浙江西比里科技有限公司与西比里新材料（新疆）有限公司共同组成。公司主营业务为电工新材料及输变电业务，主要包括电磁线的研发、生产和销售以及电线电缆、变压器、电抗器、开关柜等其他输变电产品的研发和销售。我们面向国家“双碳”战略与新型电力系统发展的重大需求，联合西安交通大学电工材料电气绝缘全国重点实验室、上海交通大学金属基复合材料全国重点实验室等科研机构，开发出高性能电工材料和电力装备系列产品，并提供新一代电-算一体高效能源解决方案。

公司致力于为客户提供专业、高效、可持续的解决方案，推动电工新材料领域的技术创新与绿色发展。通过“国际化创新集群+产业集群”的双引擎，构建"国际化创新集群+产业集群"双引擎体系，立足国际合作，以高能级创新平台实现前沿技术研发与全球资源联动，打造国际化科技产业创新集群与产业升级。

Xibili Technology is composed of Zhejiang Xibili Technology Co., Ltd., a National High-Tech Enterprise, and Xibili New Materials (Xinjiang) Co., Ltd. The company focuses on electrical new materials and power transmission and transformation business, including the R&D, production, and sales of magnet wires, as well as the development and sales of other power transmission and transformation products such as wires and cables, transformers, reactors, and switchgear.

In response to national "dual-carbon" goals and the development needs of new power systems, we collaborate with leading research institutions: the State Key Laboratory of Electrical Insulation and Power Equipment and the State Key Laboratory of Metal Matrix Composites to develop high-performance electrical materials and power equipment. We also provide next-generation integrated electro-computing energy-efficient solutions.

Dedicated to delivering professional, efficient, and sustainable solutions, we continuously drive technological innovation and green development in the field of electrical new materials. Through the dual-engine of "International Innovation Clusters + Industrial Clusters," we leverage international cooperation and high-level innovation platforms to advance cutting-edge R&D and global resource integration, fostering an international technology industry innovation ecosystem and industrial upgrading.

2019	100+	2+4
成立时间 Foundation	员工人数 100+ Employees	2大生产基地，4大研发平台 2 production bases, 4 R&D

西比里科技-产业布局
Industrial Layout

乌鲁木齐，Urumqi

西比里新材料(新疆）有限公司
中俄（新疆）国际实验室

西安，Xi'an

西安交通大学
电工材料电气绝缘全国重点实验室

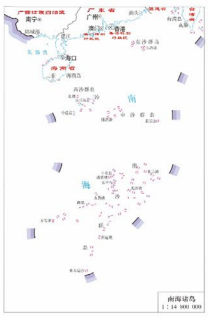
昆山，Kunshan

西比里新材料（苏州）有限公司

诸暨，Zhuji

浙江西比里科技有限公司
中俄（诸暨）国际实验室





授权专利 106
Granted Patents

国际专利 8
PCT

符合全球认证体系
Compliant with global certification systems

ISO9001

主要产品-电工材料
Core Products-Electrical Materials

铝合金电线电缆 Aluminum Wire and Cable



性能特点 Performance Characteristics

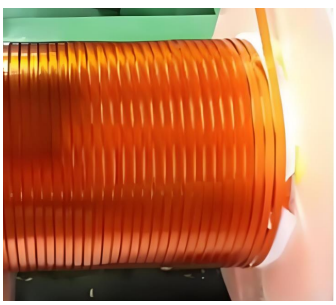
- 重量为铜导体的45%
  45% weight reduction
- 耐低温、耐腐蚀、防火、防水
  Low-Temperature, Corrosion, Fire, and Water Resistance
- 可靠的绝缘与防护
  Reliable insulation and protection
- 导体面积较铜上升150%，采用压紧线芯，外径仅增大11%
  Conductor Area: 150% greater than copper. Overall Diameter: Only 11% larger, achieved through a compacted core design

应用领域 Applications

- 电力、汽车、交通、矿产、船舶、化工、军事等领域
  Widely used in sectors such as Power, Automotive, Transportation, Mining, Marine, Chemical Industry, and Military
- 各类复杂恶劣的工程环境
  Complex and hostile engineering environments

铝基电磁线/箔 Aluminum Electromagnetic Wire/Foil







性能特点 Performance Characteristics

无机陶瓷绝缘
Inorganic Ceramic Insulation

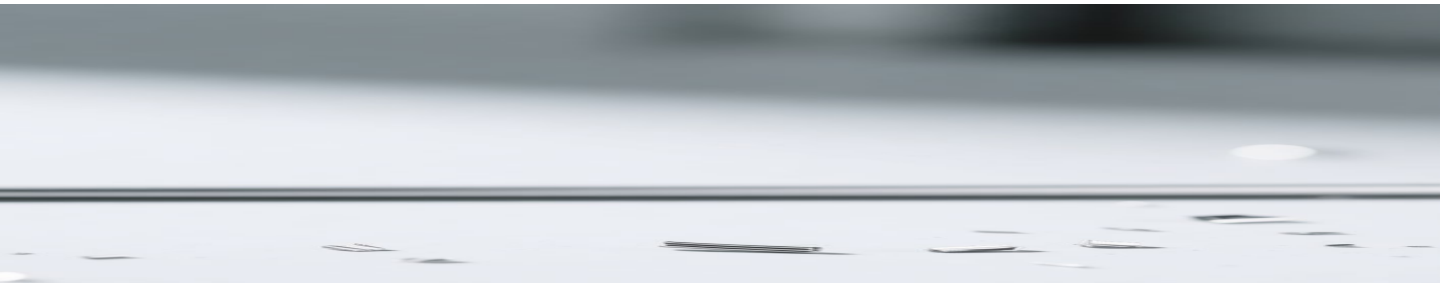
- 至高300℃绝缘耐温
  Heat Resistance: Up to 300℃
- 10 W/(m·K)高导热系数
  High Thermal Conductivity
- 厚度可调抗弯折
  Customizable-Thickness & Flexible

无机+有机复合绝缘
Inorganic-Organic Hybrid Insulation

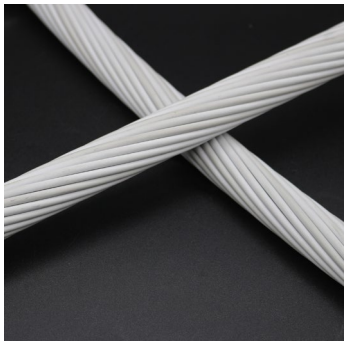
- 耐宽频高压、耐电晕
  Excellent resistance to wide-frequency high voltage and corona
- 优异的抗局放长寿命
  Excellent PD Resistance, Long Service Life
- 较传统电气性能提升至多提升50%
  Boosts electrical performance by up to 50% versus conventional materials

应用领域 Applications

- 高温环境电机、电器驱动器
  Motors and Electrical Drives in High-Temperature Environments
- 油浸式配电变压器
  Oil-Immersed Distribution Transformers
- 铝导体箔绕变压器
  Aluminum Conductor Foil-Wound Transformers
- 风力发电机绕组
  Wind Turbine Generator Stators
- 干式变压器/电抗器
  Dry-Type Transformers / Reactors
- 新能源充电桩
  EV Charging Piles
- 牵引变压器/电机
  EV Charging Piles



铝合金架空线 Aluminum Overhead Lines



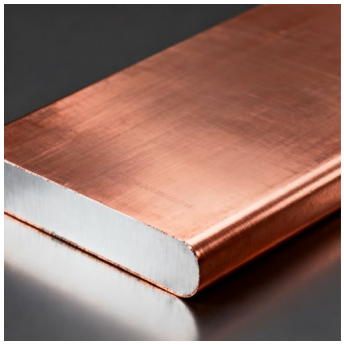
性能特点 Performance Characteristics

- 减少电晕损耗
  Reduced Corona Loss
- 增加机械强度
  Enhanced Mechanical Strength
- 耐风沙、耐紫外线、耐腐蚀
  Resistance to Sand, UV, and Corrosion
- 延缓结冰、加速融冰
  Delayed Ice Accumulation and Accelerated De-icing

应用领域 Applications

- 高压输电线路
  High-Voltage Transmission Lines
- 城市电网主干网络
  Urban Grid Main Networks
- 跨区域电力互联工程
  Inter-regional Power Interconnection

铜铝复合板/带/箔 Copper-Clad Aluminum Composite Plate, Sheet, and Foil



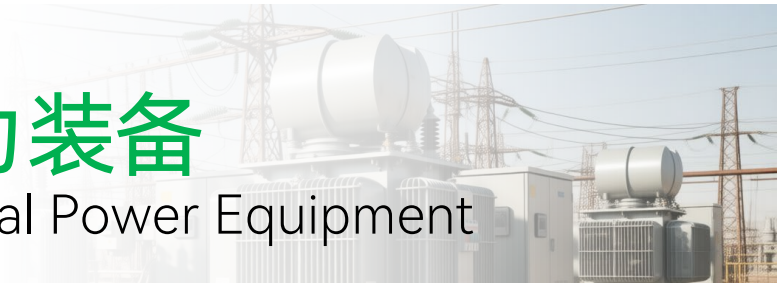


性能特点 Performance Characteristics

- 成本约为纯铜排的65%
  cost savings of up to 35% compared to a pure copper busbar
- 载流量约为纯铜排的85%-87%
  85%-87% of the current capacity of a solid copper busbar
- 纯铜(T2)与纯铝复合而成，安全有保障
  Fabricated as a composite material from pure copper (T2) and pure aluminum, safety guaranteed
- 可靠的冶金结合，结合强度超过铝材自身强度
  Reliable metallurgical bond with superior integrity

应用领域 Applications

- 母线槽
  Busway
- 变压器接电排
  Transformer Busbar
- 佛手夹、开关柜(成套设备)
  Palm Grip Clamp, Switchgear (Complete Set of Equipment)
- 储能模块搭接排
  Energy Storage Module Busbar
- 充电桩接地排、汇流排、电气连接件
  Charging Pile Grounding Busbar, Current Collector, Electrical Connector
- 电气冲压件
  Electrical Stamping Part
- 替代H62黄铜，T2紫铜，无氧铜
  Replaces H62 Brass, T2 Red Copper, and Oxygen-Free Copper



主要产品-电力装备
Core Products-Electrical Power Equipment

陶瓷复合绝缘变压器/电抗器 CERACO™ Transformer/ Reactor



性能特点 Performance Characteristics

干式变压器/电抗器
Dry-Type Transformer/Reactor

- 整机重量减轻15-20%
  The overall unit is 15-20% lighter
- 优异的过载能力提供备用容量
  Superior Overload Capability with Built-in Spare Capacity
- 防开裂，长寿命，免维护
  Crack Resistance, Extended Lifespan, Maintenance-Free



油浸式配电变压器
Oil-Immersed Distribution Transformers

- 低粘度、高闪点绝缘油
  Low-Viscosity, High-Flash-Point Oil
- 更低导体界面热阻，热传导更快
  Superior Thermal Performance
- 低损耗、运行安静、环境友好
  Low Loss, Quiet Operation, Environmentally Friendly

应用领域 Applications

- 风力发电机舱变
  Wind turbine nacelle transformer
- 新能源发电配电
  Power Distribution Center
- 用户侧配电中心
  End-User Power Distribution Center
- 偏远恶劣工况环境
  Remote and Harsh Operating Environments
- 主干电网变电站
  Main Grid Substation
- 大容量风电/光伏升压站
  Large-Scale Renewable Energy (Wind/Solar) Step-Up Substation

新能源箱式变压器 Pad-Mounted Transformer for New Energy Power



性能特点 Performance Characteristics

- 高能效
  High Energy Efficiency
- 全天候可靠运行
  All-Weather Reliable Operation
- 快速部署光伏/风电/储能/充电站
  Rapid Deployment for PV/Wind/Storage/Charging Stations
- 数字化接入，助力智慧能源管理
  Digital Integration for Smart Energy Management

应用领域 Applications

- 新能源发电升压站
  New Energy Generation Step-Up Substation
- 电化学储能系统
  Battery Energy Storage System
- 电动汽车超级充电站
  EV Supercharging Station
- 分布式能源微电网
  Microgrid



环保气体中低压金属封闭开关柜
Eco-Gas Insulated Medium- and Low-Voltage Metal-Enclosed Switchgear



性能特点
Performance Characteristics

- 模块化设计
  Modular Design
- 全封闭防护结构
  Fully Enclosed Protective Structure
- 智能监测集成(温度、局放等实时监测)
  Integrated Intelligent Monitoring (Real-time monitoring of temperature, partial discharge, etc.)
- 高可靠性元器件，降低维护成本
  High-reliability components, reduced maintenance costs

应用领域
Applications

- 电力系统、石油化工、冶金制造、公共事业
  Serving Key Sectors: Power Systems, Petrochemical, Metallurgy, and Public Utilities
- 配电系统的电力分配、转换、控制和无功功率补偿
  Distributing, converting, and controlling power systems, along with providing reactive power compensation

可调脉冲电源 Tunable Pulse Power Supply



性能特点
Performance Characteristics

- 系统采用模块化设计、集成度高
  A modular design with a high degree of integration
- 双脉冲输出(频率和占空比连续可调)
  Dual-pulse output (frequency and duty cycle are continuously adjustable)
- 风冷，产线安装方便，环境要求低
  Cooled by forced air, allowing for easy production line installation and presenting low environmental requirements
- 保护电路完善故障率低，产品维护成本及维护周期短，低成本
  With comprehensive protection circuits and a low failure rate, the product offers low maintenance costs, short maintenance cycles, and low overall cost of ownership

应用领域
Applications

- 微弧氧化等表面处理
  Micro-arc Oxidation and Other Surface Treatments
- 各类污水处理(脱盐除外)
  Various Types of Wastewater Treatment (Excluding Desalination)





铝合金电线电缆

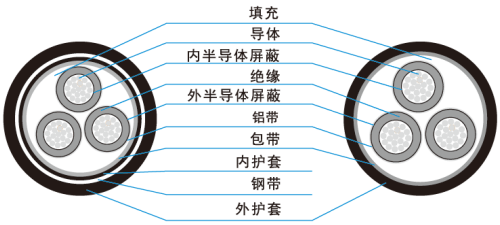
Aluminum Wire and Cable

铝芯聚氯乙烯绝缘电缆(450/750 V及以下)
Aluminum core PVC insulated cable (450 / 750 V and below)

标称截面 Nominal section (mm^2)	导体中单线最少根数 Minimum number of single wires in a conductor	绝缘厚度 Insulation thickness (mm)	平均外径上限 Upper limit of average outer diameter (mm)	20℃时导体直流电阻最大值 Maximum DC resistance of conductor at 20℃ (Ω/km)
2.5	1	0.8	3.9	12.1
4	1	0.8	4.4	7.41
6	1	0.8	5.0	4.61
10	7	1.0	6.7	3.08
16	7	1.0	7.8	1.91
25	7	1.2	9.7	1.20
35	7	1.2	10.9	0.868
50	19	1.4	12.8	0.641
70	19	1.4	14.6	0.443
95	19	1.6	17.1	0.320

电缆电线符合(Cable Compliance) :GB/T 5023-2008《额定电压450/750V及以下聚氯乙烯绝缘电缆》

挤包绝缘电力电缆(1-35 kV)
Extruded insulated power cables (1-35 kV)



电缆结构示意图
Cable Structure Diagram

高绝缘性能 High Insulation Performance
高质量绝缘材料(交联聚乙烯) 保证高电压下安全运行

高机械强度 High Mechanical Strength
足够的机械强度满足直接埋地、穿管、架空等安装方式

高耐腐蚀性 High Corrosion Resistance
外护套具有耐候性、耐油、耐化学腐蚀特性

阻水结构 Water-Blocking Structure
可以有效防止水分渗透到电缆内部，保护电缆免受水分影响

防火阻燃 Flame Retardancy
具备一定的阻燃性能，以降低火灾风险

型号 Model	产品名称 Product Name	产品范围 Production Range
X(A、B、C、D)-YJLHV	铝芯交联聚乙烯绝缘聚氯乙烯护套(A、B、C、D级阻燃)电力电缆 XLPE Insulated PVC Sheathed Aluminum Conductor Power Cable (with Flammability Ratings A, B, C, D)	芯数：1芯 Number of Cores: 1 Core 截面 Section: 25-630 mm^2 芯数：2-3芯 Number of Cores: 2-3 Cores 截面 Section: 25-400 mm^2
X(A、B、C、D)-YJLHV22	铝芯交联聚乙烯绝缘钢带铠装聚氯乙烯 XLPE Insulated Steel Tape Armored, PVC Sheathed Aluminum Conductor	
X(A、B、C、D)-YJLHV23	铝芯交联聚乙烯绝缘钢带铠装聚氯乙烯 XLPE Insulated Steel Tape Armored, PVC Sheathed Aluminum Conductor	
X(A、B、C、D)-YJLHV32	铝芯交联聚乙烯绝缘细圆钢丝铠装聚氯乙烯护套(A、B、C、D级阻燃)电力电缆 XLPE Insulated, Fine Round Steel Wire Armored, PVC Sheathed Aluminum Conductor Power Cable (with Flammability Ratings A, B, C, D)	
X(A、B、C、D)-YJLHV42	铝芯交联聚乙烯绝缘粗圆钢丝铠装聚氯乙烯护套(A、B、C、D级阻燃)电力电缆 XLPE Insulated, Large Round Steel Wire Armored, PVC Sheathed Aluminum Conductor Power Cable (with Flammability Ratings A, B, C, D)	

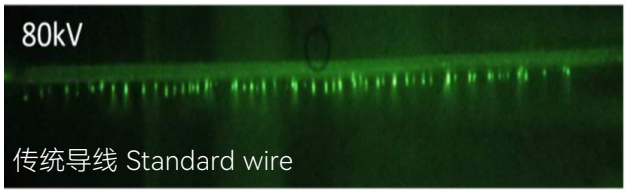
电力电缆符合(Cable Compliance) : GB/T 12706-2020《额定电压1kV到35kV挤包绝缘电力电缆及其附件》

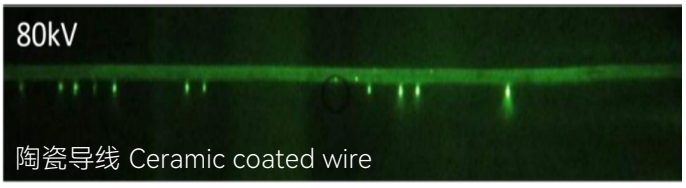
铝合金架空线

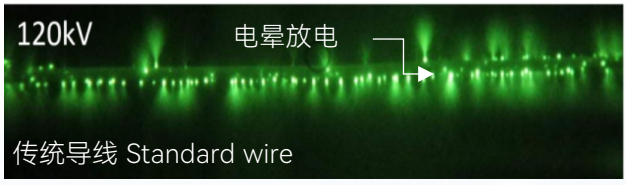
Aluminum Overhead Lines

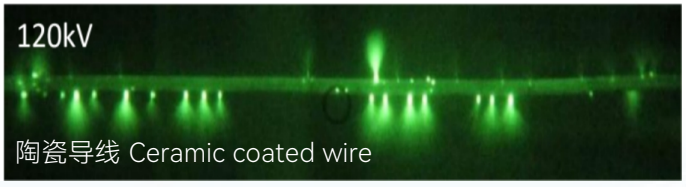
本公司可提供采用陶瓷化处理后的铝绞线和钢芯铝绞线，使其在所有型号上具有天然耐候特性，有效减少线路中的电晕损耗（最多降低40%）。特别适用于高海拔、沿海及重污染区域输电线路。抗覆冰型产品能有效延缓覆冰速率，减少冰层厚度并提升融冰速度。产品兼具节能与强环境适应性，是传统铝及铝合金架空线的理想升级方案。

Our purpose-engineered, ceramic-coated aluminum and ACSR conductors offer built-in weather resistance for all models, reducing corona losses by up to 40%. They excel in harsh environments—from high altitudes to coasts and polluted areas—while our anti-icing versions slow icing, limit buildup, and speed up melting. This combination of efficiency and durability makes them the definitive upgrade from standard aluminum and aluminum alloy conductors.

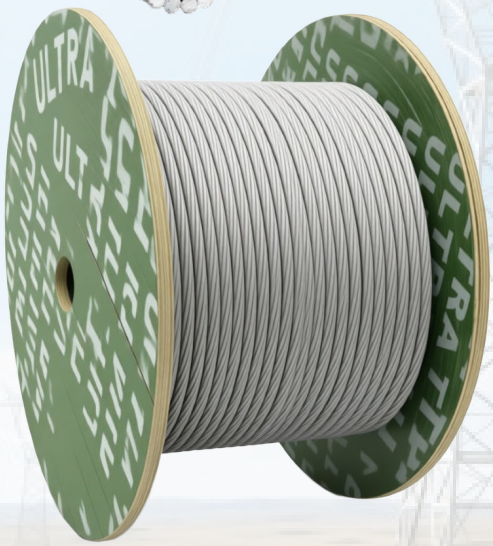












陶瓷铝绞线
Aluminum Stranded Wire

型号(Type)：JTCL
截面(Section)： 10-630 mm^2

陶瓷铝合金绞线
Aluminum Alloy Stranded Wire

型号(Type)：JTCLHA1、 JTCLHA2
截面(Section)： 20-720 mm^2

钢芯陶瓷铝合金绞线
Aluminum Alloy Conductor Steel Reinforced

型号(Type)：JTCLHA1/G1A、 JTCLHA2/G1A、JTCLHA1/G1B、 JTCLHA2/G1B、JTCLHA1/G3A、JTCLHA2/G3A、
铝/钢面积 aluminum/steel area： 18/3-725/30 mm^2

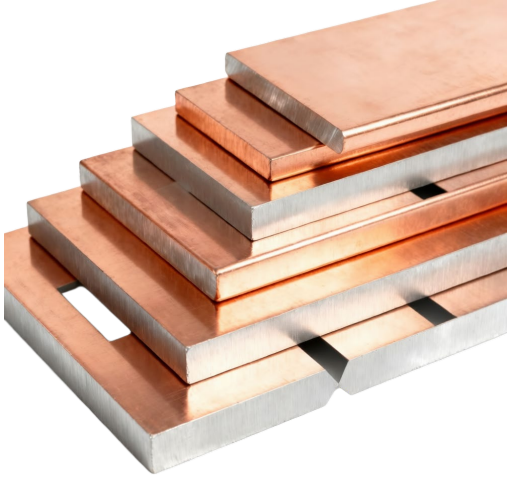
架空线符合(Cable Compliance) :GB/T 1179-2008《圆线同心绞架空导线》



铝基电磁线/箔

Aluminum Based Magnet Wire/ Foil

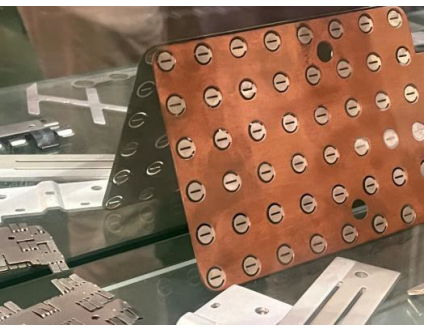
类型 Category	产品名称 Product Name	型号 Type	耐热等级 Thermal Class	性能特点 Performance Characteristics	生产范围 Dimensions
陶瓷电磁线/箔 Ceramic Magnet Wire / Foil	陶瓷铝扁(圆）线 Ceramic Aluminum Flat Wire	TX/B(Y)/300	300	耐高温、高导热、耐热冲击性好，机械强度好 High Thermal Conductivity, Excellent Thermal Shock Resistance, Good Mechanical Strength	扁线规格: Flat wire: 2- 42mm² 圆线规格: Round wire: 0.1-3.5 mm
	陶瓷铝箔A型 Ceramic Aluminum Foil Type A	TBA/300		为复合结构方案提供优异的界面结合基础 A superb base for interfacial adhesion, enabling robust composite structures	宽度(Width): 3-100 mm 厚度 (Thickness): 0.085-0.3 mm
	陶瓷铝箔B型 Ceramic Aluminum Foil Type B	TBB/300		有效应对高频屈服效应，减少铝箔边缘局部放电 Effectively Manages the High-Frequency Yield Effect, Mitigates Edge Partial Discharge	
陶瓷复合绕包线 Ceramic Composite Wrapped Wire	聚酯薄膜绕包 陶瓷复合绝缘铝扁(圆）线 Polyester-Film-Wrapped, Ceramic-composite Insulated Aluminum Flat/Round Wire	CMZ/B(Y)/155	155	常规环境和消费级绝缘产品 Standard environments and consumer-grade products	扁线规格: Flat wire: 2- 42mm² 圆线规格: Round wire: 0.1-3.5 mm
	聚酯薄膜无纺布绕包 陶瓷复合绝缘铝扁(圆）线 Polyester-Film and Non-Woven-Fabric- Wrapped, Ceramic-composite Insulated Aluminum Flat Wire	CMW/B(Y)/155	155	无纺布提供额外机械保护 Providing additional mechanical protection	
	玻璃丝绕包 陶瓷复合绝缘铝扁(圆）线 Glass-Fiber-Wrapped, Ceramic-Composite Insulated Aluminum Flat/Round Wire	CMB/B(Y)/180	180	优异的耐高温、机械性能 High-temperature resistance, mechanical properties	
	聚酰亚胺薄膜绕包 陶瓷复合绝缘耐电晕铝扁(圆）线 Polyimide-Film-Double-Wrapped, Corona-Resistant Ceramic composite Insulated, Aluminum Flat/Round Wire	CMY/B(Y)/200 CMY/B(Y)/220 CMY/B(Y)/240	200-240	卓越的电气和耐高温性能，更长的耐电晕寿命 Superior electrical and thermal performance and extended corona-resistance	
陶瓷复合漆包线 Ceramic Composite Enameled Wire	聚酯漆包 陶瓷复合绝缘铝扁(圆）线 Polyester-Enameled, Ceramic-Composite Insulated Aluminum Flat/Round Wire	CQZB(Y)/130	130	出色的机械性能、优良的电气绝缘性、低成本 Outstanding mechanical properties, excellent electrical insulation, and low cost	
	聚酯聚酰亚胺漆包 陶瓷复合绝缘铝扁(圆）线 Polyester-Enameled, Ceramic-Composite Insulated Aluminum Flat/Round Wire	CQZYB(Y)/200	200		
	聚酰亚胺漆包 陶瓷复合绝缘铝扁(圆）线 Polyimide-Enameled, Ceramic-Composite Insulated Aluminum Flat/Round Wire	CQYB/200 CQYB/220 CQYB/240	200-240	优异的耐高低温性能，超强结合力、优异的化学稳定性和阻燃性 Exceptional resistance to high and low temperatures, extremely strong bonding, outstanding chemical stability, and excellent flame retardancy	



铜铝复合板/带/箔

Copper-Clad Aluminum Composite Plate, Sheet, and Foil

铜层截面积比例 20%，铜层质量比重 45%
Copper by cross-sectional area: 20%
Copper by mass: 45%





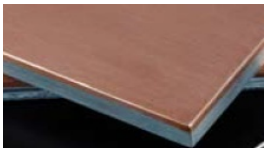




复合母线产品 Composite Busbar

宽度 Width：20 – 140 mm

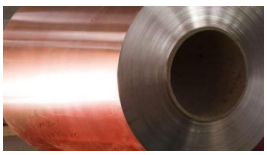
厚度 Thickness：3 – 20 mm



复合箔带产品 Composite Foil Strip

宽度 Width：20 – 130 mm

厚度 Thickness：0.5 – 3 mm



技术指标 Specifications

规格尺寸 Dimensions

直流电阻率 DC Resistivity：≤0.0248 Ω mm^2/m

载流量：相当于纯铜排的 85%—87%
Ampacity: 85% to 87% compared to a pure copper busbar.

延伸率 Elongation：12%—35%

抗拉强度 Tensile Strength：≥110 N/mm

密度 Density：3.94 g/cm^3

屈服应力 Yield Stress：44 MPa—54 MPa

杨氏模量 Young's Modulus：71 GPa

导热系数 Thermal Conductivity：0.58 Cal/(cm·s·℃)











陶瓷复合绝缘变压器/电抗器/箱式变压器

CERACO™ / CERAFOL™ Transformer/ Reactor/ Pad-Mounted Transformer

新型弹性容量技术承载额外能源需求
Novel Overload Capacity Technology Supports Fluctuating Energy Needs

1 采用双侧绝缘、低损耗的晶粒取向电工钢片制成的三柱式铁芯
Three-limb core made of grain-oriented, low-loss electric sheet steel insulated on both sides

2 低压绕组使用铝带绕制，其线匝通过预浸绝缘片进行永久性粘结固定
Low-voltage winding made of aluminum strip; turns are permanently bonded with insulating sheet

高压绕组由采用陶瓷复合箔绕或线绕技术的单个铝线圈制成，并通过真空浇注工艺完成
High-voltage winding made of individual aluminum coils using CERACO™/CERAFOL™ technology and vacuum casting

3 外绝缘系统采用环氧树脂/石英粉混合物，这使得变压器基本免维护、防潮、适用于热带地区，并且具有耐火和自熄特性
Insulation made of an epoxy resin/ quartz powder mixture makes the transformer extensively maintenance free, moisture-proof and suitable for the tropics, fire-resistant, and self extinguishing

铝绕组与环氧树脂具有相近的热膨胀系数，减少温升带来的开裂与局部放电，获得更长寿命
Thanks to the closely matched thermal expansion coefficients of the aluminum winding and epoxy resin, mechanical stress from temperature swings is minimized. This effectively prevents cracking and partial discharge, delivering a longer operational lifespan.

安全与效益兼顾
Safety and Benefits

轻量化与高安全性，高波动负载适应性
Lightweight & Safe: Ensures reliable adaptation to highly fluctuating loads

成本与环保平衡，有效降低电站的初始投资
Cost-Effective & Eco-Conscious: Significantly reduces the CAPEX

坚固耐用，免维护，高能效
Durable & Efficient: Delivers maintenance-free operation with high energy efficiency

低压绕组温度监测配备PTC热敏电阻探测器(PT100传感器可选)
Temperature monitoring with PTC thermistor detector in the low-voltage windings (alternatively: Pt100 sensors)

可选变压器局放监测VS30-V或UHF传感器，具体选型需结合变压器结构、安装空间及监测系统兼容性
Optional PD monitoring with VS30-V or UHF sensors, depending on transformer's construction, spatial constraints, and monitoring system compatibility

气候等级 Climatic Class：C3*
环境等级 Environmental Class：E3*
火灾分类 Fire Classification：F1*
*可根据要求提供其他气候与环境等级
*Further climate and environment classes on request



型号系列

Product Series

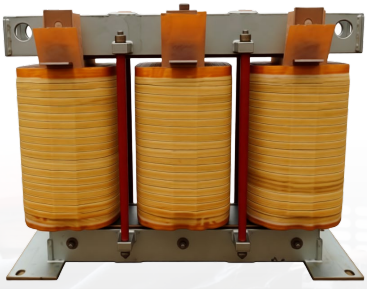
电压等级 Voltage level：10 kV、20 kV、35 kV、66 kV、110 kV、220 kV	
额定容量范围 Rated capacity range：30-10000kVA	高压绕组：铝导体；低压绕组：铝带 High-voltage: aluminum conductor; low-voltage: aluminum strip
最高绝缘等级 Highest insulation class：C级	冷却方式：自然空气冷却/强迫风冷 Cooling: natural air / forced air
能效等级 Energy efficiency class：1/2/3级	结构形式：环氧树脂真空浇注/开敞式 Construction type: Epoxy resin vacuum casting / Open-Vented



陶瓷绝缘开敞式干式变压器系列
Ceramic-Insulated Open-Vented Dry-Type Transformer Series



陶瓷膜包复合绝缘开敞式干式变压器系列
Ceramic-Film Insulated Open-Vented Dry-Type Transformer



陶瓷纸包复合绝缘干式电抗器系列
Ceramic-Paper Insulated Dry-Type Reactor Series



陶瓷绝缘油浸式配电变压器系列
Ceramic-Insulated Oil-Immersed Distribution Transformer Series



新能源箱式变压器
Pad-Mounted Transformer for New Energy Power

应用案例
Application Cases

1. 高原耐候型陶瓷复合干式变压器投运应用(新疆邦特电器制造有限公司)
2. 沙漠戈壁箱式变电站投运应用(新疆西部晶源电力建设有限公司)
3. 新型陶瓷铝导线及其在油浸式配电变压器中的应用(新疆中泰特种电力设备有限公司)
4. 江苏昆山花溪小学变压器
5. 陶瓷膜包复合绝缘干式变压器用于扩容改建工程(浙江西比里科技有限公司)

环保气体中低压金属封闭开关柜

Eco-Gas Insulated Medium- and Low-Voltage Metal-Enclosed Switchgear

本系列产品采用干燥空气、氮气或低GWP氟化混合气体（如g^3、HFO/CO_2混合气）作为主绝缘介质，替代传统SF_6，实现零或近零温室效应。设备为模块化金属封闭结构，适用于单母线、单母线分段等主接线方式，具备高可靠性、免维护及环境适应性强等特点，广泛应用于新能源电站、城市配网、轨道交通、工矿企业及数据中心等场景。

This product series employs dry air, nitrogen, or low-GWP fluorinated gas mixtures (e.g., g^3, HFO/CO_2 blends) as the primary insulating medium, replacing traditional SF_6 to achieve zero or near-zero greenhouse effect. Designed with a modular, metal-enclosed structure, it is suitable for single-busbar and single-busbar sectioning layouts. Characterized by high reliability, maintenance-free operation, and strong environmental adaptability, it is widely deployed in renewable energy plants, urban distribution networks, rail transit systems, industrial and mining facilities, and data centers.

参数类别 Parameter	参数范围 Parameter Range
额定电压 Rated Voltage	12 kV ~ 40.5 kV
额定频率 Rated Frequency	50 Hz / 60 Hz
额定电流 Rated Current	630 A ~ 3150 A
额定短时耐受电流 Rated Short-Time Withstand Current	16 kA ~ 31.5 kA / 3s ~ 4s
额定峰值耐受电流 Rated Peak Withstand Current	40 kA ~ 80 kA
工频耐受电压（1min）Power-Frequency Withstand Voltage	28 kV ~ 95 kV（对地/相间）
雷电冲击耐受电压 Lightning Impulse Withstand Voltage	75 kV ~ 185 kV
绝缘介质 Insulating Medium	干燥空气 / N_2 / 氟化腈混合气（GWP < 1）
防护等级 Ingress Protection (IP) Rating	IP4X（外壳） / IP2X（内部隔室）
内部电弧等级 Internal Arc Classification	AFLR / IAC AB级，耐受31.5kA/0.5s ~ 1s
机械寿命 Mechanical Endurance	断路器：≥10,000 次；三工位开关：≥3,000 次

环境适应性说明
Environmental Adaptability Specification

环境温度：周围空气温度不超过 +40℃，不低于 -15℃，且在 24 小时内测得的平均值不超过 +35℃；
Ambient Temperature: The surrounding air temperature shall not exceed +40℃ and shall not be lower than -15℃, with the daily average not exceeding +35℃ over any 24-hour period;

环境湿度：日平均相对湿度95%以下，月平均相对湿度90%以下；
Ambient Humidity: The average daily relative humidity is below 95%, and the average monthly relative humidity is below 90%;

开关柜安装场所海拔不超过1000米；
Installation Altitude: The switchgear installation site altitude does not exceed 1000 meters;

无火灾、无爆炸危险、无严重污秽、无化学腐蚀气体及剧烈震动的场所；
Site Conditions: The location shall be free from fire and explosion hazards, severe pollution, chemical corrosive gases, and excessive vibration;

当使用条件与上述工作条件不符时，由用户向设备提供商提出协商解决。
Non-Standard Conditions: If the actual service conditions differ from those specified above, the user shall consult with the equipment supplier for a resolution.





可调脉冲电源

Tunable Pulse Power Supply

技术参数 Specifications

交流输入电压 AC Input Voltage	三相四线 380 V±10%、50 Hz±2% Three-phase Four-wire system
额定输入相电流 Rated Input Phase Current	1-500 A
额定输出功率 Rated Output Power	40-300 kW
功率因数 Power Factor	≥0.95
峰值效率 Peak Efficiency	≥91%
输出模式 Output Mode	正负向脉冲模式、双脉冲、直流模式 Bidirectional Pulse Mode, Double Pulse, and DC Mode.
正向脉冲电压调节范围 Forward Pulse Voltage Adjustment Range	50 V ~ 800 V
负向脉冲电压调节范围 Reverse Pulse Voltage Adjustment Range	-800 V ~ -50 V
电流调节范围 Current Adjustment Range	1.5-200 A (单向)
输出频率调节范围 Output Frequency Adjustment Range	正负向脉冲 Bidirectional Pulse： 100 Hz -6000 Hz 双脉冲 Double Pulse：50 Hz-3000 Hz
占空比可调范围 Duty Cycle Adjustment Range	5% ~ 95%

可靠性 Reliability

软启动功能；欠压、过压、缺相、过流保护； 整流模块超温保护，风扇故障报警；功率模块温度过高报警、超温保护；输出短路、输出绝缘低、过压、过流保护。
A soft-start function and comprehensive protections, including: Input/Output: Under-voltage, over-voltage, over-current, and phase loss.
Output: Short-circuit and low insulation resistance.
Modules: Over-temperature protection and overtemperature pre-alarm for both rectifier and power modules. Auxiliary: Fan failure alarm.

MBTF ≥ 20000h

工作环境 Operating Environment

海拔高度不超过 1000 米；环境温度为-20--+40℃；空气相对湿度不超过 65%，无凝露；无剧烈震动和冲击场合，垂直倾斜度≤5°。
Designed to operate under the following conditions: at an altitude not exceeding 1000 meters; within an ambient temperature range of -20 to +40°C; with relative air humidity not exceeding 65% and no condensation; in locations free from severe vibration and impact, and with a vertical inclination of ≤5°.

定制化 Bespoke

按您的需求，从电压、接口到散热方案，我们的模块化电源系统，为您的应用场景量身打造，无缝集成。
With our modular power systems, everything from voltage and interfaces to thermal management is bespoke, engineered to integrate flawlessly into your unique application.

污水处理-高效节能电破乳
Advanced Energy-Efficient Electrolytic Demulsification for Sewage Treatment



新疆吉庆页岩油压裂返排液水净化
Purification of Shale Oil Fracturing Flowback Fluid from Jiqing, Xinjiang

能耗 Energy Consumption：传统电化学的1/5-1/2，吨水能耗0.005 ~ 5 kW
Energy consumption is reduced to 1/5 to 1/2 of that required by traditional electrochemical methods, a net energy usage of 0.005 to 5 kW per ton of water treated.
适应性 Adaptability：处理效果明显，污泥产量为絮凝、气浮的1/5-1/3
Dramatically lower sludge production, only 1/5 to 1/3 of the amount generated by conventional flocculation or air flotation processes.
处理助剂 Additives：不使用任何助剂 None
停留时间 Hydraulic Retention Time：2–200 min
智能性 Intelligent Features：自动区分污染成分和水，使电能主要作用于污染成分，大幅提高电能利用率。 The system automatically differentiates between pollutants and water, ensuring that electrical energy primarily targets the pollutants, which greatly enhances power utilization efficiency.
压降 Pressure Drop：≈0

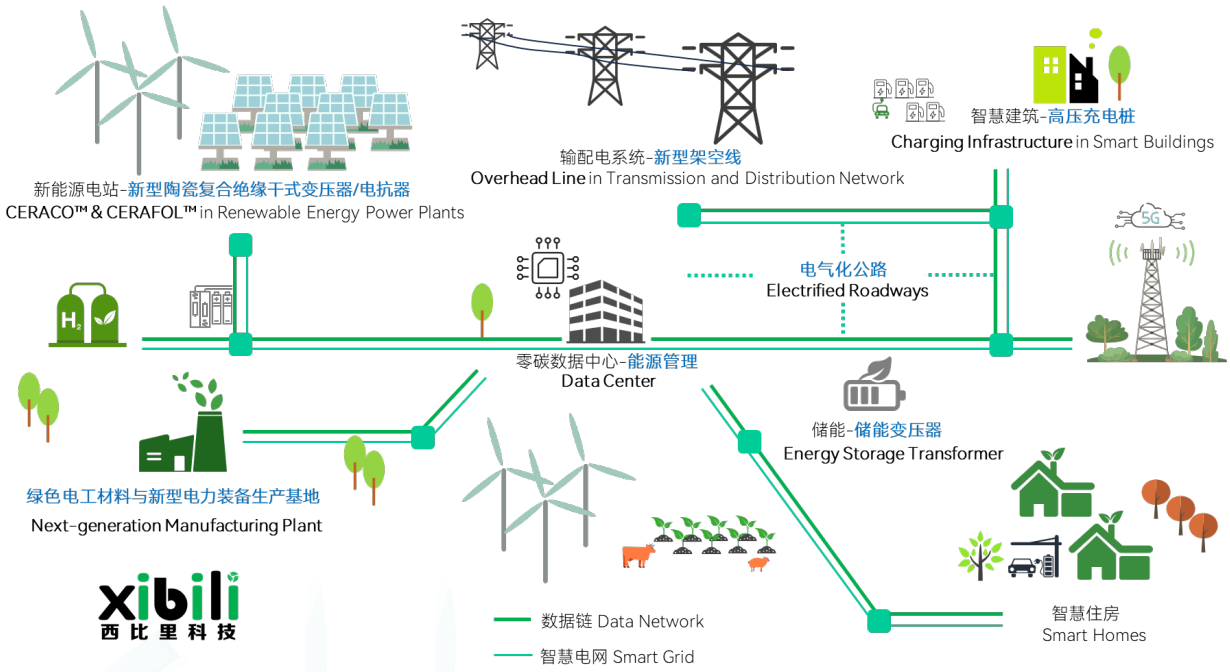


新型电力-系统解决方案
Dependable Power Grid

建设可靠有韧性的电网 Robust & Resilient

展望2035年，全球电力消费将从2024年的28万亿千瓦时跃升至40万亿千瓦时，电力生产体系亦将迎来历史性重构，人工智能革命也正以前所未有的广度与深度重塑人类文明，其背后是指数级增长的算力需求与能源消耗。超级智能数据中心单体功耗已达数十亿瓦量级，未来5至10年，全球电力基础设施的承载能力与能源供给韧性，将成为决定AI演进速度与智能文明高度的核心瓶颈。在此背景下，以高比例可再生能源为主体、深度融合数字智能技术的新型电力系统，正崛起为支撑智能时代的“能源神经网络”——不仅输送电力，更承载算力、联结智能、驱动未来。

By 2035, global electricity consumption is projected to surge from 28 trillion kilowatt-hours in 2024 to 40 trillion kilowatt-hours. The power generation system will undergo a historic transformation, while the artificial intelligence revolution is reshaping human civilization with unprecedented breadth and depth—driven by exponentially growing demands for computing power and energy. Today, the power consumption of a single hyperscale intelligent data center has reached the gigawatt level. Over the next 5 to 10 years, the capacity and resilience of global power infrastructure and energy supply will become the core bottlenecks determining the pace of AI evolution and the height of intelligent civilization. Against this backdrop, a new type of power system—dominated by high proportions of renewable energy and deeply integrated with digital and intelligent technologies—is emerging as the “energy neural network” of the intelligent era: not only transmitting electricity, but also carrying computing power, connecting intelligence, and driving the future.



一体化协调保障优质与稳定电力供应

Integrated and coordinated mechanisms ensure high-quality and stable power supply



然而，风光能源固有的间歇性与波动性，对电力系统的稳定性、灵活性与韧性提出了前所未有的挑战。为此，我们率先提出并践行“**新材料—新装备—新系统**”三位一体的创新范式，以底层材料科学突破驱动高端电力装备革新，以装备升级赋能系统级能源架构重构，最终打造“**电-算一体、绿智融合**”的新一代电力直送体系。这一范式不仅为智能世界提供绿色、可靠、高效的能源底座，更将深度重构全球制造业的价值链与技术生态，引领企业迈向高质量、高效率、高可持续性的新质生产力时代。

However, the inherent intermittency and volatility of wind and solar energy pose unprecedented challenges to the stability, flexibility, and resilience of power systems. In response, we have pioneered and implemented an innovative “Trinity Paradigm” of New Materials – New Equipment – New Systems. This approach drives breakthroughs in advanced power equipment through fundamental innovations in materials science, empowers system-level energy architecture transformation through equipment upgrades, and ultimately builds a next-generation integrated power–computing direct transmission system that seamlessly combines green energy with intelligent technology.

This paradigm not only provides a green, reliable, and efficient energy foundation for the intelligent world, but also profoundly restructures the global manufacturing value chain and technological ecosystem—leading enterprises into a new era of productivity characterized by high quality, high efficiency, and high sustainability.

合作伙伴
Cooperative Partner

上海交通大学 金属基复合材料全国重点实验室 THE STATE KEY LAB OF METAL MATRIX COMPOSITES

西安交通大学 XI'AN JIAOTONG UNIVERSITY | 电工材料电气绝缘全国重点实验室 State key laboratory of electrical insulation and power equipment

西安交通大学 XI'AN JIAOTONG UNIVERSITY

CORE Advancing Knowledge for Humanity

CHINAGAS 中國燃氣

The State Scientific Institution “THE JOINT INSTITUTE OF MECHANICAL ENGINEERING OF THE NATIONAL ACADEMY OF SCIENCES OF BELARUS”

中车永济电机有限公司 CRRC YONGJI ELECTRIC CO., LTD.

Новосибирский государственный университет *НАСТОЯЩАЯ НАУКА

2019 MHCK

中国南方电网 CHINA SOUTHERN POWER GRID

明珠电气 PEARL ELECTRIC

中国电气装备集团有限公司 China Electrical Equipment Group Co., Ltd.

СО РАН 俄罗斯科学院西伯利亚分院

国家电网公司 STATE GRID CORPORATION OF CHINA

排名不分先后 No Particular Order
谨此向刊登及未刊登的合作伙伴表示诚挚的谢意!
Thanks for all of the listed and unlisted partners!

行业影响
Impact

能源 Energy

新装备构筑非碳清洁能源
New equipment builds a Non-Carbon Clean Energy

装备 Equipment

材料革新驱动新装备发展
Materials Innovation Drives the Development of New Equipment

材料 Material

综合性能更好的可循环材料
Recyclable Material with Superior Overall Performance









与时俱增 与时俱进



“一带一路”科技合作

Belt and Road Initiative Science and Technology Cooperation



受邀出席阿塞拜疆国家科学院80周年庆典
Guest Invitation, 80th Anniversary of the Azerbaijan National Academy of Sciences

国际欧亚实验室合作协议
International Eurasian Laboratory Cooperation Agreement



STRATEGIC COOPERATION AGREEMENT
ON THE ESTABLISHMENT OF THE INTERNATIONAL EURASIAN LABORATORY
(THE “IEL”)
BETWEEN
INTERNATIONAL CORE ACADEMY OF SCIENCES AND HUMANITIES (“CORE ACADEMY”)
AND
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